

CSCI3230 (ESTR3108)

Fundamentals of Artificial Intelligence

Lecture 3. Logistic Regression

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Outline

Part 1. Classification problem

Part 2. Logistic regression for classification

Part 3. Optimizing logistic regression

Part 4. Summary



Part 1. Classification problem

Example: spam detection

Automatically detect whether a received email is spam.

Greetings to you. [Scam](#)

★ Mrs Ruth Bader G <*****@xxxx.com>

Be careful with this message. Similar messages were used to steal people's personal information.

Reply-To: xxxxx@XXX.com

Bcc: foolidiot@XXX.com

[Reply](#) | [Reply to all](#) | [Forward](#) | [Print](#) | [Delete](#) | [Show original](#)

Greetings to you.

My name is Mrs Ruth Bader G Cooks from the USA. I know that this message might come to you as a surprise because we don't know each other nor have we ever met before, but accept it with an open and positive mind. I have a Very important request that made me to contact you;

I was diagnosed with ovarian cancer, which doctors have confirmed and announced to me that I have just a few days to leave, Now that I'm ending the race like this, without any family members and no child, I just came across your email contact from my personal search.

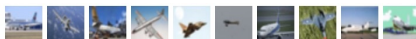
I'm a business woman from the USA dealing with gold exportation here in the Republic of Burkina Faso. I have decided to hand over the sum of (\$3.5 Million Dollar) in my account to you for the help of orphanage homes/the needy once in your location to fulfill my wish on earth. But before handing over my data's to you, kindly assure me that you will take only 50% of the money and share the rest to orphanage homes/the needy once in your country. Return to enable me forward to you the bank contact details now that I have access to Internet in the hospital to enable you contact the bank, always check your e-mail, and always remember me for doing good.

Please don't forget to reply to me in my Private E-mail:() Your early response will be appreciated.

Example: image classification

Classify the content of images. Cat? Dog? Ship? ...

airplane



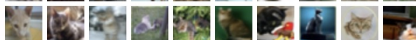
automobile



bird



cat



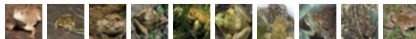
deer



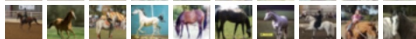
dog



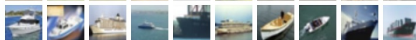
frog



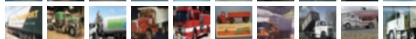
horse



ship

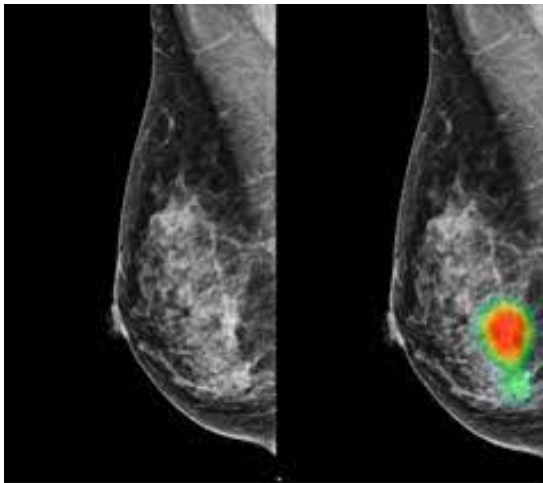


truck



Example: cancer detection

Diagnose medical images. Malignant tumor? or Benign lesion?



Basic concepts

What is classification?

¹<https://en.wikipedia.org/wiki/Classification>

What is classification?

- Classification is a process related to categorization, in which ideas and objects are recognized, differentiated and understood. ¹

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- **Goal:** Given a set of samples, with different categories, learn a classifier. Then, the classifier can automatically recognize new samples within the set of given categories.

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Compare classification with regression.

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Compare classification with regression.

- The output of classification is a discrete value.
- The output of regression is a continuous value.

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Notations

- Samples (e.g. images or texts) are represented as feature vectors.
- Suppose we have m training samples, each has n -dimensional features. We can write these samples as:

$$X^{(1)} = (x_1^{(1)}, \dots, x_n^{(1)})$$

$$X^{(2)} = (x_1^{(2)}, \dots, x_n^{(2)})$$

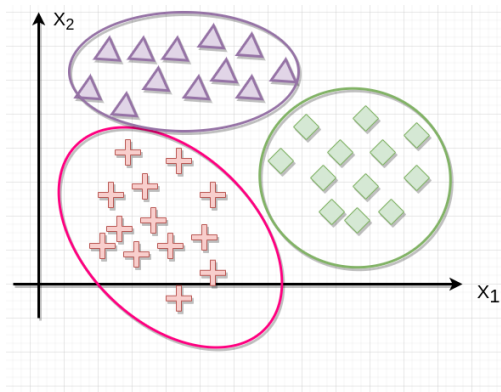
$$\vdots$$

$$X^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)})$$

with their corresponding category labels as $y^{(1)}, y^{(2)}, \dots, y^{(m)}$.

Classification

- The figure below gives an illustration of classification for samples with 2-dimensional features.
- Similar to finding a way to separate samples to different parts.



Binary vs. multi-class classification

Binary classification

- Only two categories, usually, use 0, 1 to label the two categories. (0 is to label negative samples and 1 is to label positive samples.)
- Examples:

Binary vs. multi-class classification

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 - Whether an email is spam or normal? (0 for spam, 1 for normal email)
 - Whether a photo contains a cat? (0 for no, 1 for yes)
 - Whether you will be admitted to CUHK?

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Multi-class classification

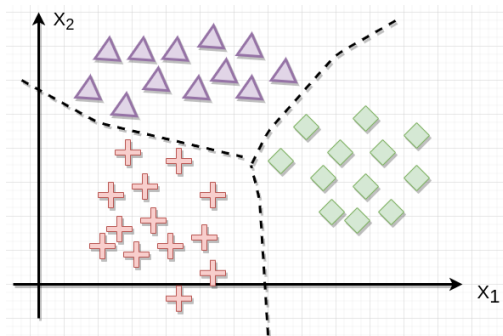
- More than two categories, usually, use 1,2,3,... to label categories.
- Examples:
 - Which animal is in the photo, cat / dog / pig / duck?
 - Which CUHK College is a student associated with, S.H. Ho, New Asia, Chung Chi, Morningside, ...?

Decision boundary

- Usually consider partitioning the entire feature space into multiple parts, one for each category (a.k.a. **class**). Data points associated to the same category lie in the same partition.

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- Intuitively, **decision boundary** is a union of curves or surfaces that partition the feature space.



Linear classifier

What is linear classifier?

Linear classifier

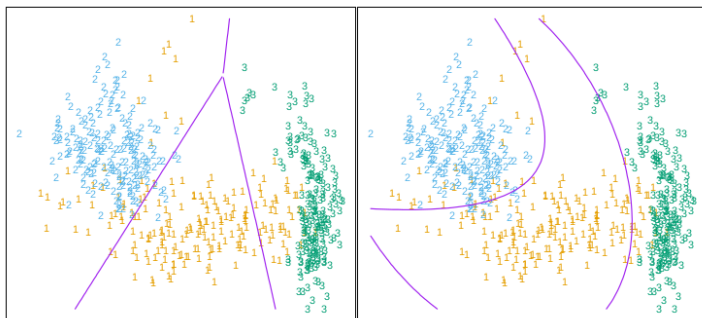
What is linear classifier?

- Linear classifier is a kind of linear models, which classifies data based on a linear combination of input features.

Linear classifier

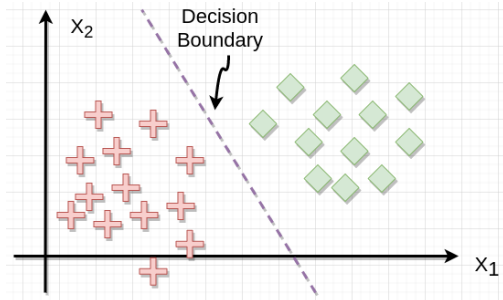
What is linear classifier?

- Linear classifier is a kind of linear models, which classifies data based on a linear combination of input features.
- Decision boundary of linear classifiers is a union of straight lines / hyperplanes, e.g., left figure.



Geometry of linear binary classifier

For a linear binary classifier, its decision boundary is a straight line / hyperplane that partitions the feature space into two half-spaces.

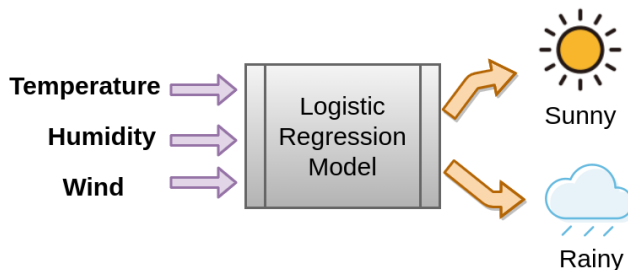




Part 2. Logistic regression for classification

Logistic regression

- Logistic regression models the **probabilities** for classification problems with two possible outcomes.
- It is an easily interpretable classification method that gives the probability of an event occurring.



Formulation of binary classification

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- For simplification, we consider linear binary classification.
- Given an input n -dimensional feature $X \in \mathbb{R}^n$, and its possible category $y \in \{0, 1\}$.
- By definition, for binary classification, we would like to construct a model to estimate $P(\hat{y} = 1|X)$, which is the probability that our prediction is 1 by giving X .
- Since only two classes in total, we have:
$$P(\hat{y} = 0|X) + P(\hat{y} = 1|X) = 1$$
- If $P(\hat{y} = 1|X) \geq P(\hat{y} = 0|X)$, we can say that $y = 1$ is more possible to happen than $y = 0$, so the sample with feature X is predicted to label 1 (i.e., $\hat{y} = 1$). Otherwise, the sample is labelled 0 (i.e., $\hat{y} = 0$).

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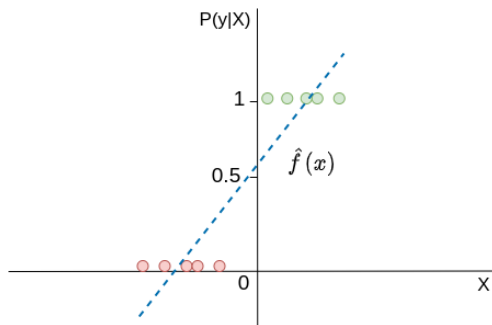
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- Equivalently, $P(\hat{y} = 1|X) \geq 0.5$, the sample should be classified as 1, otherwise, classified as 0.

Is linear regression applicable for binary classification?

- We have studied linear regression models for fitting data points.
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- We have studied linear regression models for fitting data points. Is linear regression applicable for binary classification?
- Suppose we have a set of samples:
 $\{(x_1^{(1)}, \dots, x_n^{(1)}, y^{(1)}), \dots, (x_1^{(m)}, \dots, x_n^{(m)}, y^{(m)})\}$, where $y^{(i)} \in \{0, 1\}$.
- How about fitting those data points by linear regression such that $\hat{f}(X)$ predicts the **probability** of $P(\hat{y} = 1|X)$?

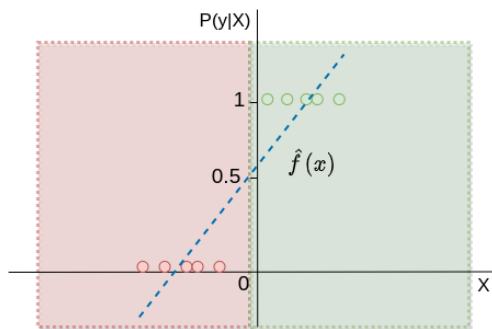


Is linear regression applicable for binary classification?

- Unfortunately, the range of linear functions is $(-\infty, +\infty)$, which is not suitable to output probability.

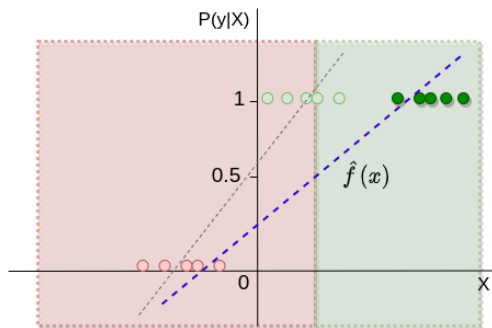
Is linear regression applicable for binary classification?

- Unfortunately, the range of linear functions is $(-\infty, +\infty)$, which is not suitable to output probability.
- What if we apply a cut-off rule, for example,
 - If $\hat{f}(X) \geq 0.5$, $\hat{y} = 1$
 - If $\hat{f}(X) < 0.5$, $\hat{y} = 0$



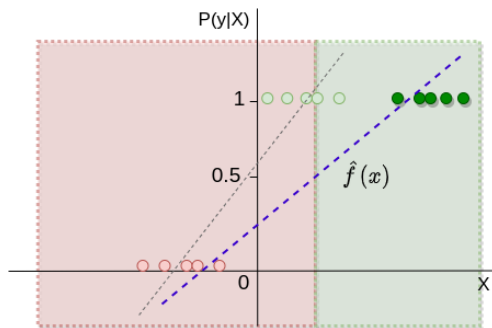
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- When there are some outliers, linear regression with cut-off may fail to produce considerable results.



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- However, linear regression with cut-off rule would be very sensitive to those outliers.
- When there are some outliers, linear regression with cut-off may fail to produce considerable results.
- It is hard to directly use linear regression with cut-off to classification!



Probability and odds

The cut-off rule is a transformation which maps linear functions' range $(-\infty, +\infty)$ to the range of probability $(0, 1)$.

Can we find a better transformation for classification?

- Let's begin with odds.

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Can we find a better transformation for classification?

- Let's begin with odds.
- Event A and its probability to happen: $P(A) = p, P(A^C) = 1 - p$.
- The odds of event A is defined as

$$\text{odds}(p) = \frac{p}{1 - p}$$

(Note: the odds of an event is the ratio of the probability of an event to the probability of its complement.)

Probability and odds

Examples of odds:

- ① Suppose we flip a coin, the probability of landing on head is 0.5. Equivalently, the odds of landing on head is 1 to 1, i.e.,
 $odds = \frac{0.5}{1-0.5} = 1.$

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Equivalently, the odds of landing on head is 1 to 1, i.e.,
$$odds = \frac{0.5}{1-0.5} = 1.$$
- 2 Suppose the probability that Lakers win the NBA final is 0.75.
Equivalently, the odds that Lakers win the final is 3 to 1, i.e.,
$$odds = \frac{0.75}{1-0.75} = 3.$$

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- 1 Suppose we flip a coin, the probability of landing on head is 0.5. Equivalently, the odds of landing on head is 1 to 1, i.e.,
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- 2 Suppose the probability that Lakers win the NBA final is 0.75. Equivalently, the odds that Lakers win the final is 3 to 1, i.e.,
$$\text{odds} = \frac{0.75}{1-0.75} = 3.$$
- 3 Suppose the probability that your paper will be accepted by CVPR is 0.25. Equivalently, the corresponding odds is 1 to 3, i.e.,
$$\text{odds} = \frac{0.25}{1-0.25} = \frac{1}{3}.$$

Logit transformation

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- We define logit (a.k.a. log-odds) of an event A as

$$\text{logit}(p) = \ln \text{odds}(p) = \ln \left(\frac{p}{1-p} \right)$$

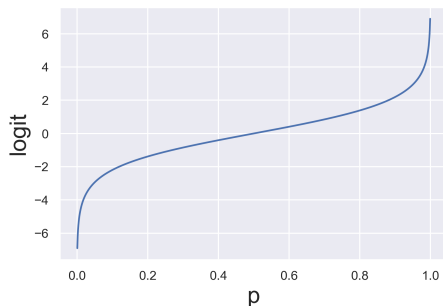
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- Inversely, in order to adopt linear functions for classification, instead of applying a cut-off rule, we can apply $\text{logit}^{-1}(\cdot)$, which transforms any value in $(-\infty, +\infty)$ to probability in $(0, 1)$.

Find $\text{logit}^{-1}(\cdot)$

- Note that $\text{logit}(p)$ is strictly monotonically increasing
 $\implies \text{logit}^{-1}(\cdot)$ must exist.

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$$z = \text{logit}(p) = \ln\left(\frac{p}{1-p}\right)$$

$$-z = \ln\left(\frac{1-p}{p}\right)$$

$$e^{-z} = \frac{1}{p} - 1$$

$$p = \frac{1}{1 + e^{-z}}$$

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$$p = \frac{1}{1 + e^{-z}}$$

$$\implies \text{logit}^{-1}(z) = p = \frac{1}{1 + e^{-z}}$$

Derivation of logistic regression

- Recall the linear function:

$$\hat{f}_{\Theta}(X) = X^T \Theta + \theta_0$$

- Recall our goal is to find $P(\hat{y} = 1|X)$.
- We apply logit transformation on the linear function to fit the probability for classification prediction:

$$P(\hat{y} = 1|X) = \text{logit}^{-1}(X^T \Theta + \theta_0) = \frac{1}{1 + e^{-(X^T \Theta + \theta_0)}}$$

The equation above is the basic formulation of **logistic regression**.

Logistic sigmoid function

Logistic sigmoid function:

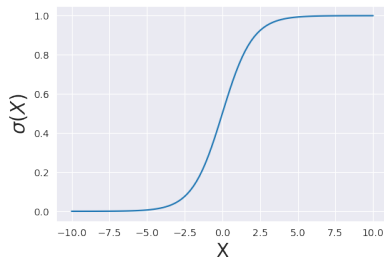
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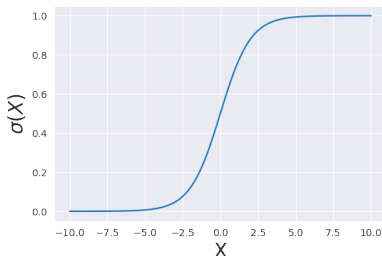


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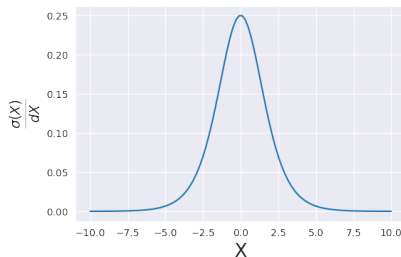
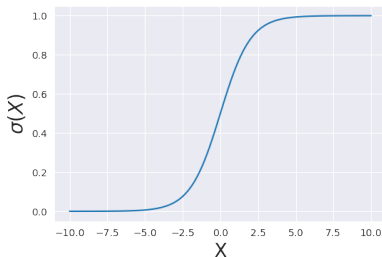


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- $\sigma(0) = 0.5, \lim_{x \rightarrow +\infty} \sigma(x) = 1, \lim_{x \rightarrow -\infty} \sigma(x) = 0$.
- $\sigma(x)$ is monotonic, and its first derivative is bell shaped.



Logistic regression for classification

Logistic regression:

$$P(\hat{y} = 1|X) = \sigma(X^T\Theta + \theta_0) = \frac{1}{1 + e^{-(X^T\Theta + \theta_0)}}$$

Logistic regression for classification

Logistic regression:

$$P(\hat{y} = 1|X) = \sigma(X^T\Theta + \theta_0) = \frac{1}{1 + e^{-(X^T\Theta + \theta_0)}}$$

- As specified before

$$P(\hat{y} = 1|X) \geq 0.5 \implies \hat{y} = 1 \text{ and } P(\hat{y} = 1|X) < 0.5 \implies \hat{y} = 0.$$

- Θ and θ_0 are parameters learned from training data.
- For simplification, we absorb the θ_0 into Θ (likewise in linear regression):

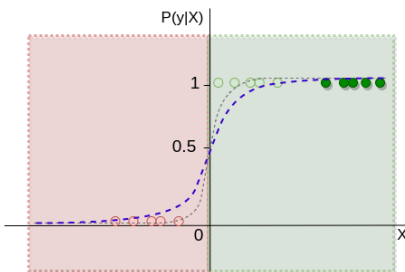
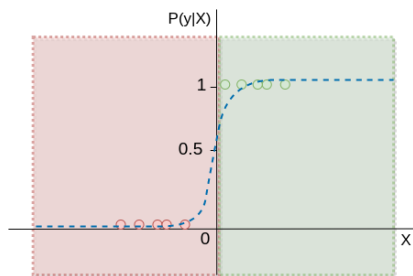
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- Logistic regression exactly outputs the probability of $y = 1$ conditioned on the known input X .
- $P(\hat{y} = 1|X)$ changes significantly around $X^T\Theta = 0$, which is a good characteristic to distinguish samples with different labels.
- Logistic regression is more robust to outliers.

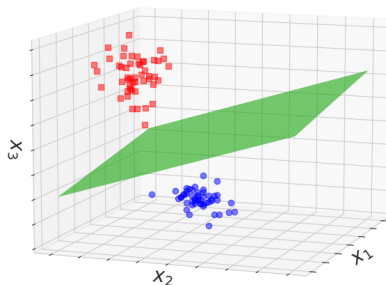
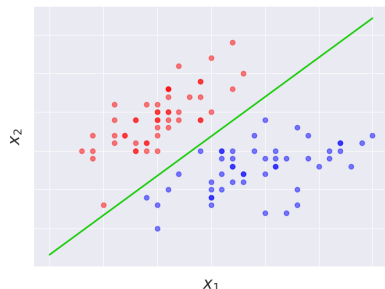


Logistic regression is a (generalized) linear classifier

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 - 1 $X^T\Theta \geq 0$: $\hat{y} = 1$
 - 2 $X^T\Theta < 0$: $\hat{y} = 0$
- The geometry of $X^T\Theta$ is a hyperplane. $X^T\Theta \geq 0$ is the space above the hyperplane, while $X^T\Theta < 0$ is the space below the hyperplane.
- The decision boundary is the hyperplane $z = X^T\Theta$.





Part 3. Optimizing logistic regression

Optimizing logistic regression

How to search for the optimal Θ through the training data to maximize the classification performance? Recall our logistic regression model:

$$P(\hat{y} = 1|X) = \sigma(X^T \Theta) = \frac{1}{1 + e^{-(X^T \Theta)}}$$

- Given that we have training data:

$$\{(x_1^{(1)}, \dots, x_n^{(1)}, y^{(1)}), \dots, (x_1^{(m)}, \dots, x_n^{(m)}, y^{(m)})\}$$

where $y^{(i)} \in \{0, 1\}$ is the given corresponding training label.

Supervision criterion

For each sample $X^{(i)} = (x_1^{(i)}, \dots, x_n^{(i)})$. We have that

$$P(\hat{y}^{(i)} = 1 | X^{(i)}) = \sigma(X^{(i)T} \Theta) = \frac{1}{1 + e^{-(X^{(i)T} \Theta)}}$$

which is function w.r.t. Θ , we learn the best Θ from the given data.

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Supervision criterion:

- If $y^{(i)} = 1$, we expect $P(\hat{y}^{(i)} = 1 | X^{(i)})$ approaches 1 as close as possible.
- If $y^{(i)} = 0$, we expect $P(\hat{y}^{(i)} = 1 | X^{(i)}) = 1 - P(\hat{y}^{(i)} = 0 | X^{(i)})$ approaches 0 as close as possible.

Supervision criterion

We can write the probability that the label is correctly predicted as:

$$p_i \stackrel{\text{def}}{=} P\left(\hat{y}^{(i)} = 1|X^{(i)}\right)^{y^{(i)}} \left(1 - P(\hat{y}^{(i)} = 1|X^{(i)})\right)^{1-y^{(i)}}$$

- When $y^{(i)} = 0 \implies p_i = P(\hat{y}^{(i)} = 0|X^{(i)})$
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- When $y^{(i)} = 1 \implies p_i = P(\hat{y}^{(i)} = 1|X^{(i)})$
- So no matter which category the sample belongs to, p_i is able to represent the probability that the prediction $\hat{y}^{(i)}$ matches the true label $y^{(i)}$.

Supervision criterion

- Next, we consider the joint probability for all the training samples, and maximize it to learn Θ .
- Since the samples are independent to each other, the joint probability can be written as:

$$\begin{aligned} p(\Theta) &\stackrel{\text{def}}{=} P(\hat{y}^{(1)} = y^{(1)}, \dots, \hat{y}^{(m)} = y^{(m)} | X^{(1)}, \dots, X^{(m)}) \\ &= P(\hat{y}^{(1)} = y^{(1)} | X^{(1)}) \dots P(\hat{y}^{(m)} = y^{(m)} | X^{(m)}) \\ &= \prod_{i=1}^m P(\hat{y}^{(i)} = 1 | X^{(i)})^{y^{(i)}} (1 - P(\hat{y}^{(i)} = 1 | X^{(i)}))^{1-y^{(i)}} \\ &= \prod_{i=1}^m p_i \end{aligned}$$

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- $p(\Theta)$ is also known as the likelihood function w.r.t. Θ .
- The method that estimates the parameters of a probabilistic model by maximizing a likelihood function is called **maximum likelihood estimation (MLE)**.

Solve MLE for logistic regression

Maximizing $p(\Theta)$ is equivalent to minimizing $-\ln p(\Theta)$ (why?).

$$\begin{aligned} E(\Theta) &= -\ln p(\Theta) = -\ln \left(\prod_{i=1}^m p_i \right) = -\sum_{i=1}^m \ln p_i \\ &= -\sum_{i=1}^m \ln \left(P(\hat{y}^{(i)} = 1 | X^{(i)})^{y^{(i)}} (1 - P(\hat{y}^{(i)} = 1 | X^{(i)}))^{1-y^{(i)}} \right) \\ &= -\sum_{i=1}^m \left[y^{(i)} \ln P(\hat{y}^{(i)} = 1 | X^{(i)}) + (1 - y^{(i)}) \ln (1 - P(\hat{y}^{(i)} = 1 | X^{(i)})) \right] \end{aligned}$$

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- In general, summation is much easier to optimize compared with product since summation operation is linear regarding to derivative.

For each single sample, we define $E_i(\Theta)$ as:

$$E_i(\Theta) = -y^{(i)} \ln P(\hat{y}^{(i)} = 1 | X^{(i)}) - (1 - y^{(i)}) \ln (1 - P(\hat{y}^{(i)} = 1 | X^{(i)}))$$

thus we rewrite $E(\Theta) = \sum_{i=1}^m E_i(\Theta)$

Cross-entropy error function

Each $E_i(\Theta)$ is exactly in the form of **cross-entropy error function**:

$$\text{cross-entropy error} = -y \ln P(y) - (1 - y) \ln (1 - P(y)), \quad y \in \{0, 1\}$$

(formulation by definition, it is a concept from information theory. $P(y)$ is the probability density function of taking value y .)

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- In our case, $P(y) = P(\hat{y}^{(i)} = 1|X^{(i)})$, which is w.r.t. Θ .

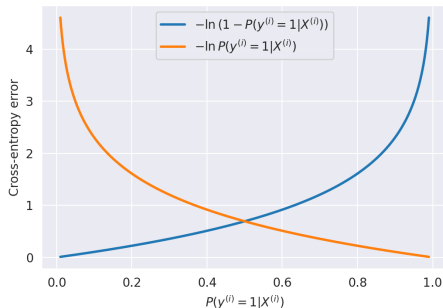
- When $y^{(i)} = 0$,

$$E_i(\Theta) = -\ln(1 - P(\hat{y}^{(i)} = 1|X^{(i)})) = -\ln\left(1 - \frac{1}{1 + e^{-X^{(i)T}\Theta}}\right)$$

- When $y^{(i)} = 1$, $E_i(\Theta) = -\ln P(\hat{y}^{(i)} = 1|X^{(i)}) = -\ln\left(\frac{1}{1 + e^{-X^{(i)T}\Theta}}\right)$

Cross-entropy error function

We plot the graph of $E_i(\Theta)$, which shows a good property of cross-entropy error function: convex! (see details in ESTR3108 notes)



In fact, sum of convex functions is also convex (linearity of convexity).

$E_i(\Theta)$'s are convex $\implies E(\Theta)$ is convex

$\implies$ global minimum = local minimum.

Minimize cross-entropy error function

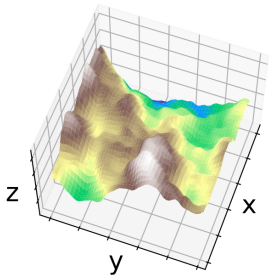
- Analogy to minimizing RSS, we would like to find $\frac{\partial E(\Theta)}{\partial \Theta} |_{\Theta^*} = 0$
- However, finding an analytic solution of $\frac{\partial E(\Theta)}{\partial \Theta} = 0$ is very tough! (You can try it by yourself 🙄)

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- However, finding an analytic solution of $\frac{\partial E(\Theta)}{\partial \Theta} = 0$ is very tough! (You can try it by yourself 😞)
- Don't be frustrated!
We have another workaround: **gradient descent**.
- Instead of finding the solution directly by mathematical tricks and reduction, gradient descent algorithm iteratively searches for the optimal solution (iteratively = step by step).

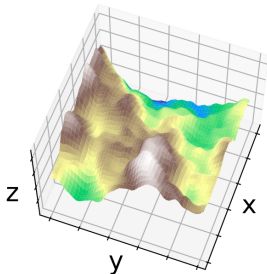
Intuition of gradient descent

- Suppose the geometry of the objective function $z = f(x, y)$ is a surface that looks like a mountain.
- Iterative optimization is similar to the scenario where we search for a path to the foot of the mountain step by step.



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- Iterative optimization is similar to the scenario where we search for a path to the foot of the mountain step by step.



Greedy scheme: For each step, we only walk downhill. To descend faster, we choose to walk along the steepest slope.

Gradient descent

Gradient descent is motivated by this greedy scheme: for each optimization step, tune parameters towards the direction where the objective function descends fastest.

Gradient descent

Gradient descent is motivated by this greedy scheme: for each optimization step, tune parameters towards the direction where the objective function descends fastest.

- Under certain conditions, gradient descent guarantees the convergence to a local minimum.
- If our objective function is convex, gradient descent is able to find the global minimum once the algorithm converges to a local minimum.

Gradient descent

So, how can we find the steepest (fastest) descent direction?

Proposition

For a twice differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient $\nabla f(\Theta)$ points in the direction of the steepest ascent or descent at Θ .

The gradient $\nabla f(\Theta)$ is a vector defined as:

$$\nabla f(\Theta) = \frac{\partial f(\Theta)}{\partial \Theta} = \left(\frac{\partial f(\Theta)}{\partial \theta_1}, \dots, \frac{\partial f(\Theta)}{\partial \theta_n} \right)$$

where θ_i is the i -th element (dimension) of Θ .

Understanding gradient descent

- As we know, derivatives imply the ascent / descent speed of functions.
- Greater directional derivative means steeper (faster) ascent / descent along a certain direction (imagine in high-dimensional space).
- By definition, the derivatives of $f(\Theta)$ along an arbitrary direction is:

$$D_{\mathbf{v}} f(\Theta) = \lim_{h \rightarrow 0} \frac{f(\Theta + h\mathbf{v}) - f(\Theta)}{h}$$

where $\mathbf{v} \in \mathbb{R}^n$, $\|\mathbf{v}\|_2 = 1$ is a unit vector along that direction.

- For all possible directions, the above proposition exactly claims that $\nabla f(\Theta)$ is the direction that maximizes the derivative of f at Θ .

Proof: $\nabla f(\Theta)$ maximizes the directional derivative

Fixed any $\Theta \in \mathbb{R}^n$, $\mathbf{v} \in \mathbb{R}^n$, $\|\mathbf{v}\|_2 = 1$, if we define $g(h)$ as:

$$g(h) = f(\Theta + h\mathbf{v})$$

The directional derivative along $\mathbf{v}$ at Θ is:

$$D_{\mathbf{v}} f(\Theta) = \lim_{h \rightarrow 0} \frac{f(\Theta + h\mathbf{v}) - f(\Theta)}{h} = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = g'(0)$$

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Meanwhile, we can compute $g'(h)$ by chain rule:

$$\begin{aligned} g'(h) &= \frac{\partial f(\Theta + h\mathbf{v})}{\partial(\Theta + h\mathbf{v})} \frac{\partial(\Theta + h\mathbf{v})}{\partial h} && \text{(chain rule)} \\ &= \nabla f(\Theta + h\mathbf{v}) \cdot \mathbf{v} \end{aligned}$$

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$$\implies D_{\mathbf{v}} f(\Theta) = g'(0) = \nabla f(\Theta) \cdot \mathbf{v}$$

So, we get the general expression of $D_{\mathbf{v}} f(\Theta)$ for any fixed vector $\mathbf{v}$.

Proof: $\nabla f(\Theta)$ maximizes the directional derivative

Further, by Cauchy-Schwarz inequality:

$$D_{\mathbf{v}} f(\Theta) = \nabla f(\Theta) \cdot \mathbf{v} \leq \|\nabla f(\Theta)\|_2 \|\mathbf{v}\|_2$$

The equality is achieved iff $\mathbf{v}$ is a scalar multiple of $\nabla f(\Theta)$.

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This means, $\nabla f(\Theta)$ points in the optimal direction of $\mathbf{v}^*$ that maximizes directional derivative $D_{\mathbf{v}} f(\Theta)$.

Gradient descent algorithm

Gradient descent is designed to update parameters in the **opposite** direction of the gradient. Through iterative updates, an optimal solution for our goal of $\min f(\Theta)$ can be attained.

Gradient descent

Ensure: $\alpha > 0$

Initialize $\Theta \leftarrow \Theta_0$ randomly

while not converge **do**

$\Theta \leftarrow \Theta - \alpha \nabla f(\Theta)$

end while

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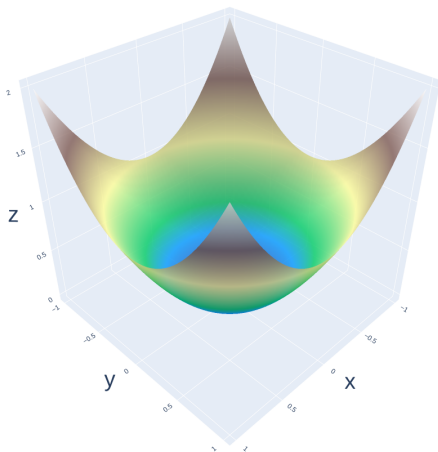
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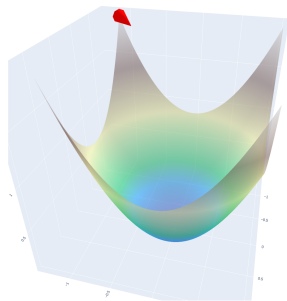
- α is a small enough hyper-parameter called **learning rate**.
 - Too small α results in very slow learning process.
 - Too large α leads to bad convergence.
 - Allowed to change at every iteration.

Example: illustration of gradient descent

Suppose to perform gradient descent on a quadratic surface $z = f(x, y) = x^2 + y^2$. Our goal is to find the optimal (x^*, y^*) such that $f(x^*, y^*)$ reaches the minimum value.

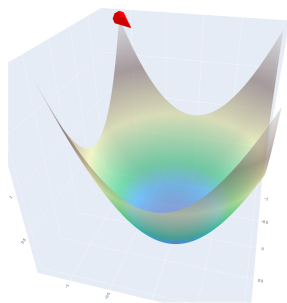


Example: illustration of gradient descent

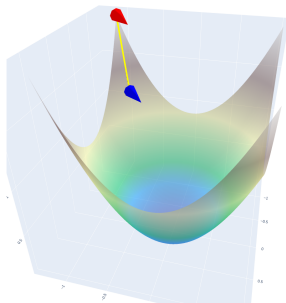


Initialize at $(-1, -1)$

Example: illustration of gradient descent

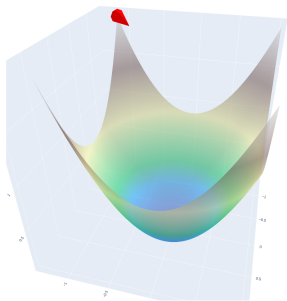


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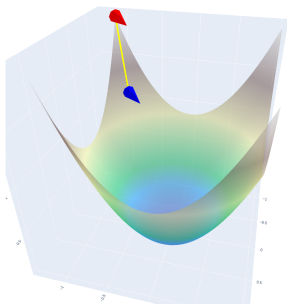


1st iteration

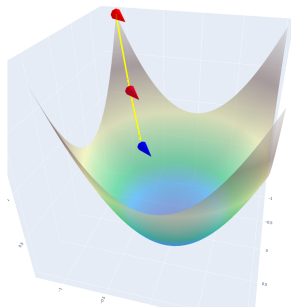
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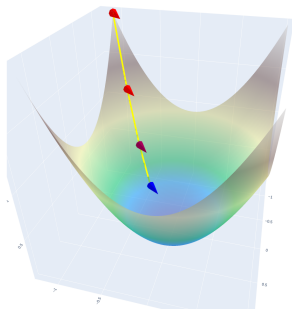


1st iteration



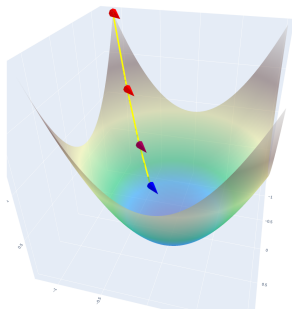
2nd iteration

Example: illustration of gradient descent

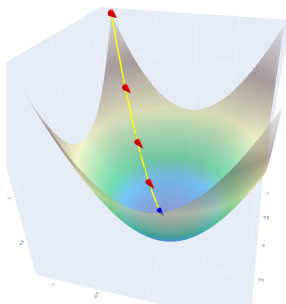


3rd iteratoin

Example: illustration of gradient descent

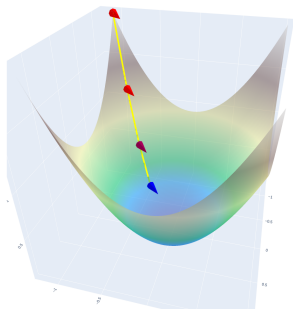


3rd iteratoin

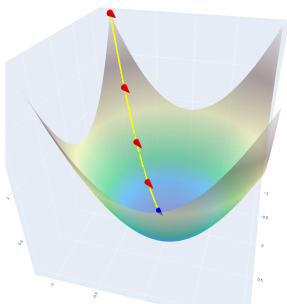


4th iteration

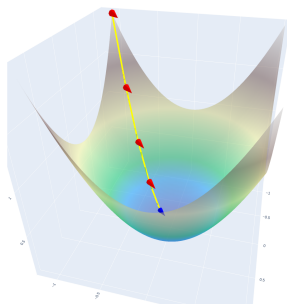
Example: illustration of gradient descent



3rd iteratoin



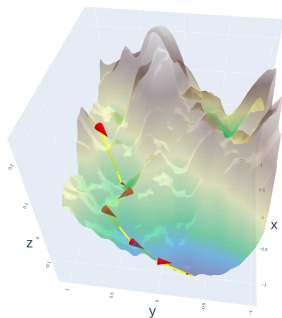
4th iteration



5th iteration: converge

Failure case of gradient descent

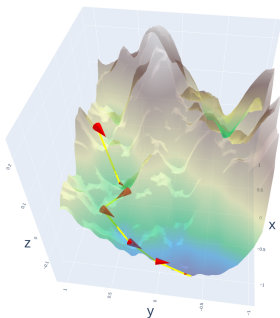
- Sometimes gradient descent also works for non-convex objective functions.



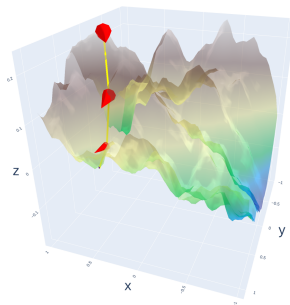
✓ Finds global minimum

Failure case of gradient descent

- Sometimes gradient descent also works for non-convex objective functions.
- But for most complex and nonconvex functions, gradient descent usually falls to a local minimum (without being able to reach the global minimum).



✓ Finds global minimum



✗ Falls to a local minimum

Find the optimal Θ for logistic regression

- Back to logistic regression
- We have defined cross-entropy error function for supervised learning of Θ from the training data.
- The analytic expression of $E(\Theta)$'s gradient:

$$\nabla E(\Theta) = \frac{\partial E(\Theta)}{\partial \Theta} = \sum_{i=1}^m \left(\frac{1}{1 + e^{-X^{(i)T} \Theta}} - y^{(i)} \right) X^{(i)}$$

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- By inserting the analytic expression to gradient descent, we finally have this algorithm to find the optimal Θ for logistic regression:

Gradient descent for logistic regression

Ensure: $\alpha > 0$

Initialize Θ randomly

while not converge **do**

$$\Theta \leftarrow \Theta - \alpha \sum_{i=1}^m \left(\frac{1}{1 + e^{-X^{(i)T} \Theta}} - y^{(i)} \right) X^{(i)}$$

end while

Quiz

Apply gradient descent to find the θ^* that minimizes $f(\theta) = \theta^2$. We set $\alpha = 0.4$, θ is initialized to 10, and regard the algorithm converges once $|f(\theta^*) - f(\theta)| \leq 0.01$. What is the number of iterations required at least to guarantee the convergence?

- A 6 iterations
- B 3 iterations
- C 4 iterations
- D not convergent

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Correct Answer: B

Quiz solution

The gradient of $\nabla f(\theta) = \frac{\partial f(\theta)}{\partial \theta} = 2\theta$. Recall the update routine: $\theta \leftarrow \theta - \alpha \nabla f(\theta)$. By taking $\alpha = 0.4$ and $\nabla f(\theta) = 2\theta$, we have

$$\theta \leftarrow \theta - 0.4 \cdot 2\theta$$

Initial $\theta_0 = 10$

Iter 1 $\theta_1 = \theta_0 - 0.8 \cdot \theta_0 = 2$, $f(\theta_1) = 4$

Iter 2 $\theta_2 = \theta_1 - 0.8 \cdot \theta_1 = 0.4$, $f(\theta_2) = 0.16$

Iter 3 $\theta_3 = \theta_2 - 0.8 \cdot \theta_2 = 0.08$, $f(\theta_3) = 0.0064$ (converges)

So the gradient descent converges at iteration 3.



Part 4. Summary

Summary

- Classification basics
- Concepts and formulation for logistic regression
- Cross-entropy error function
- Gradient descent for optimizing logistic regression

Reading materials

- Chapter 12, Advanced Data Analysis from an Elementary Point of View, Cosma Shalizi.
- Chapter 4.4, The Elements of Statistical Learning (2nd edition), T. Hastie, et al.



ESTR3108

Outline

- Explain the convexity of cross-entropy error function
- Analytic solution for logistic regression?

Convexity of cross-entropy error function

We claimed the cross-entropy error function $E_i(\Theta)$ is convex. In this part, let's go further to explain the convexity.

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Convexity of cross-entropy error function

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- Recall that

$$E_i(\Theta) = -y^{(i)} \ln P(\hat{y}^{(i)} = 1|X^{(i)}) - (1 - y^{(i)}) \ln (1 - P(\hat{y}^{(i)} = 1|X^{(i)}))$$

- For simplification, we denote $P(\hat{y}^{(i)} = 1|X^{(i)})$ as $g_i(\Theta)$:

$$g_i(\Theta) = P(\hat{y}^{(i)} = 1|X^{(i)}) = \frac{1}{1 + e^{-X^{(i)T}\Theta}}$$

- So we have $E_i(\Theta) = -y^{(i)} \ln g_i(\Theta) - (1 - y^{(i)}) \ln (1 - g_i(\Theta))$ for which we can study the convexity.

Revisit convexity

- We have studied the geometry of convex functions. How about the mathematical expression of convexity?

Revisit convexity

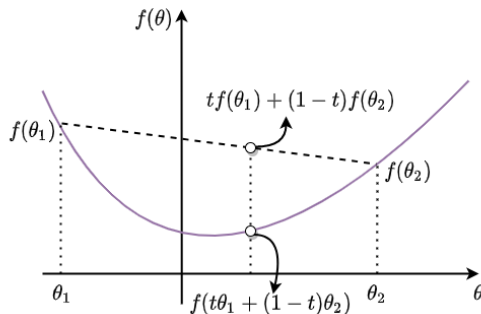
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$$f(t\Theta_1 + (1-t)\Theta_2) \leq tf(\Theta_1) + (1-t)f(\Theta_2)$$

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- We have studied the geometry of convex functions. How about the mathematical expression of convexity?
- A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff for $\forall \Theta_1, \Theta_2 \in \mathbb{R}^n, 0 \leq t \leq 1$:

$$f(t\Theta_1 + (1-t)\Theta_2) \leq tf(\Theta_1) + (1-t)f(\Theta_2)$$



Properties of convex functions

- Local minimum = global minimum.
- A twice differentiable function of one variable is convex on an interval if and only if its second derivative is non-negative there:

$$\frac{\partial^2 f(\theta)}{\partial \theta^2} \geq 0$$

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- A twice differentiable multi-variate function f is convex if and only if its Hessian matrix is everywhere positive semi-definite, which is mathematically expressed as:

$$\mathbf{H} = \frac{\partial^2 f(\Theta)}{\partial \Theta^2} \succeq \mathbf{0}$$

Convexity of cross-entropy error function

Let's find the first-order derivative of $E_i(\Theta)$:

$$E_i(\Theta) = -y^{(i)} \ln g_i(\Theta) - (1 - y^{(i)}) \ln (1 - g_i(\Theta))$$

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$$\begin{aligned} \frac{\partial E_i(\Theta)}{\partial g_i(\Theta)} &= -y^{(i)} \frac{1}{g_i(\Theta)} + (1 - y^{(i)}) \frac{1}{1 - g_i(\Theta)} \\ &= \frac{-y^{(i)} + y^{(i)} g_i(\Theta) + g_i(\Theta) - y^{(i)} g_i(\Theta)}{g_i(\Theta)(1 - g_i(\Theta))} \\ &= \frac{g_i(\Theta) - y^{(i)}}{g_i(\Theta)(1 - g_i(\Theta))} \end{aligned}$$

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Recall $g_i(\Theta)$ is a shorthand of

$$g_i(\Theta) = P(\hat{y}^{(i)} = 1 | X^{(i)}) = \frac{1}{1 + e^{-X^{(i)T} \Theta}}$$

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$$\begin{aligned}\frac{\partial g_i(\Theta)}{\partial \Theta} &= \left(-\frac{1}{(1 + e^{-X^{(i)T}\Theta})^2} \right) \left(e^{-X^{(i)T}\Theta} \right) \left(-X^{(i)} \right) \\ &= \left(\frac{1}{1 + e^{-X^{(i)T}\Theta}} \right) \left(\frac{e^{-X^{(i)T}\Theta}}{1 + e^{-X^{(i)T}\Theta}} \right) X^{(i)} \\ &= g_i(\Theta)(1 - g_i(\Theta))X^{(i)}\end{aligned}$$

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Finally, we can compute the first-order derivative:

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Convexity of cross-entropy error function

Let's find the second derivative of $E_i(\Theta)$:

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Previously, we obtained:

$$\frac{\partial E_i(\Theta)}{\partial \Theta} = (g_i(\Theta) - y^{(i)})X^{(i)}$$

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Convexity of cross-entropy error function

We also have:

$$\frac{\partial g_i(\Theta)}{\partial \Theta} = g_i(\Theta)(1 - g_i(\Theta))X^{(i)}$$

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Next step, we need to verify whether $\mathbf{H}$ is positive semi-definite.

Positive semi-definite matrix

Let's recall the definition of positive semi-definite matrix:

Definition

Symmetric matrix $A \in \mathbb{R}^{n \times n} \succeq 0$, i.e., positive semi-definite, if and only if

$$v^T A v \geq 0, \quad \forall v \in \mathbb{R}^n$$

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❷ $X^{(i)}X^{(i)T}$ is symmetric. For any v ,

$$v^T X^{(i)}X^{(i)T} v = (X^{(i)T} v)^T X^{(i)T} v = \|X^{(i)T} v\|_2^2 \geq 0$$

❸ Hence, we can conclude $\mathbf{H}$ is positive semi-definite.

Convexity of $E(\Theta)$

- Perfect, $E_i(\Theta)$ is convex, and hereby $E(\Theta)$ is convex.
- Since $E(\Theta)$ is sum of $E_i(\Theta)$'s, if each $E_i(\Theta)$ is convex, $E(\Theta)$ is convex (the sum of convex functions is also convex).

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- Proof: sum of two convex functions is a convex function
Any two convex functions $f_1(\Theta)$, $f_2(\Theta)$: $\mathbb{R}^n \rightarrow \mathbb{R}$. We have:

$$\forall 0 \leq t \leq 1 \quad f_i(t\Theta_1 + (1-t)\Theta_2) \leq tf_i(\Theta_1) + (1-t)f_i(\Theta_2) \quad i \in \{1, 2\}$$

$$\begin{aligned} f_1(t\Theta_1 + (1-t)\Theta_2) + f_2(t\Theta_1 + (1-t)\Theta_2) &\leq tf_1(\Theta_1) + (1-t)f_1(\Theta_2) \\ &\quad + tf_2(\Theta_1) + (1-t)f_2(\Theta_2) \\ (f_1 + f_2)(t\Theta_1 + (1-t)\Theta_2) &\leq t(f_1 + f_2)(\Theta_1) \\ &\quad + (1-t)(f_1 + f_2)(\Theta_2) \end{aligned}$$

$\implies (f_1 + f_2)(\Theta)$ is convex.

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$$\begin{aligned} 0 &= \frac{\partial E(\Theta)}{\partial \Theta} = \frac{\partial \sum_{i=1}^m E_i(\Theta)}{\partial \Theta} \\ &= \sum_{i=1}^m \frac{\partial E_i(\Theta)}{\partial \Theta} = \sum_{i=1}^m (g_i(\Theta) - y^{(i)}) X^{(i)} \\ &= \sum_{i=1}^m \left(\frac{1}{1 + e^{-X^{(i)T} \Theta}} - y^{(i)} \right) X^{(i)} \end{aligned}$$

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- If $E(\Theta)$ is convex, can we find the optimal Θ^* through the same way of minimizing RSS?
- Let's try to find the optimal solution Θ^* that vanishes the first-order derivative of $E(\Theta)$.

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It seems we cannot find an explicit solution to this equation...